

Name:

Student ID:

Signature:

Bonus Question [2 Marks]

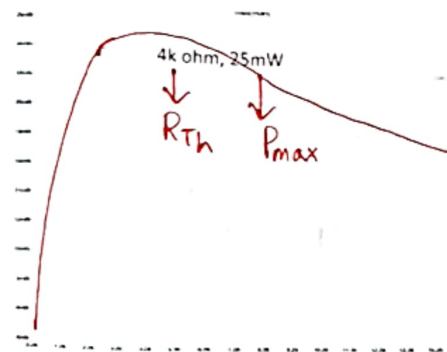
Can you apply Thevenin theorem to a circuit that contains the following elements- ac voltage source, resistor, diode, and capacitor. Why?

No. Because diode is a non-linear element.

Question 1 of 2 [8 Marks]

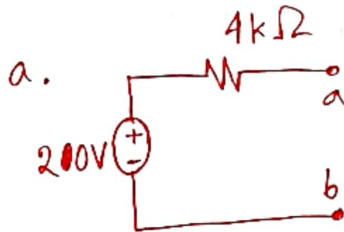
The P vs R graph shows power consumed by a load resistor (R_L).

- Draw the Thevenin equivalent circuit that produced the graph.
- Calculate efficiency for $R_L = R_{Th}$, $R_{Th}/2$, $2 \cdot R_{Th}$. Which one results in maximum efficiency?



$$P_{max} = \frac{V_{Th}^2}{4 R_{Th}}$$

$$\Rightarrow V_{Th} = 200V$$



$$b. \quad \eta = \frac{P_{out}}{P_{in}} \times 100\%$$

$$P_{in} = I^2 (R_{Th} + R_L)$$

$$P_{out} = I^2 R_L$$

$$\therefore \eta = \frac{R_L}{R_{Th} + R_L} \times 100\%$$

$$R_L = R_{Th} \Rightarrow \eta_1 = 50\%$$

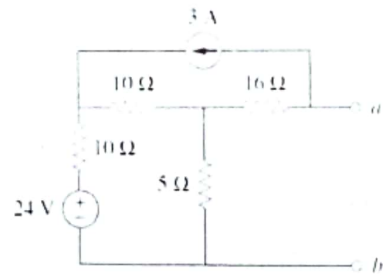
$$R_L = \frac{R_{Th}}{2} \Rightarrow \eta_2 = 33.33\%$$

$$R_L = 2R_{Th} \Rightarrow \eta_3 = 66.67\%$$

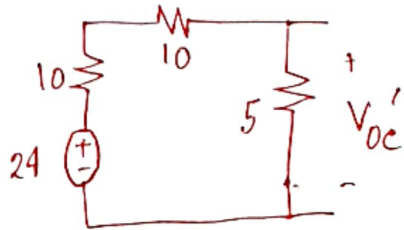
max for $R_L = 2R_{Th}$

Question 2 of 2 [12 Marks]

- a. Use superposition theorem to find the open circuit voltage between terminal a and b.
b. Draw the Thevenin equivalent circuit as seen from terminal a and b.



a. 24V active :

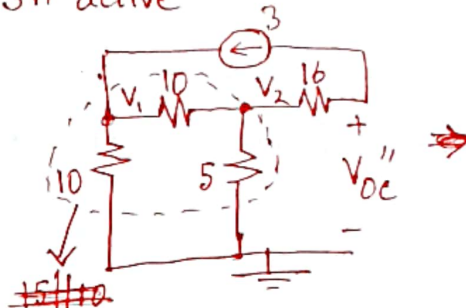


$$V'_{oc} = \frac{5}{5+10+10} \times 24 = 4.8V$$

$$\therefore V_{oc} = V'_{oc} + V''_{oc}$$

$$= \boxed{-49.2V}$$

3A active



$$V_1 \left(\frac{1}{10} + \frac{1}{10} \right) - \frac{V_2}{10} - 3 = 0$$

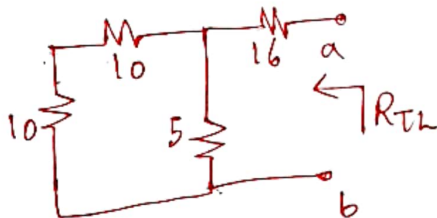
$$V_2 \left(\frac{1}{10} + \frac{1}{5} \right) - \frac{V_1}{10} + 3 = 0$$

$$\Rightarrow V_1 = 12V, V_2 = -6V$$

$$V_2 - V'_{oc} = 16 \times 3$$

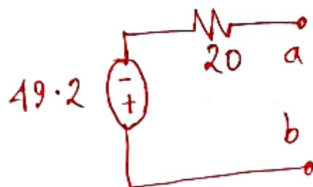
$$\Rightarrow V'_{oc} = -54V$$

b. For R_{Th} ,



$$R_{Th} = \{(10+10) \parallel 5\} + 16$$

$$= 20\Omega$$



or,



Semester: Fall 2022
 Course No: CSE250
 Course Title: CIRCUITS AND ELECTRONICS
 Section: 07

Quiz 3
 Full Marks: 20
 Time: 30 Minutes
 Date: 28 November, 2022

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Bonus Question [2 Marks]

Can you apply Thevenin theorem to a circuit that contains the following elements- ac current source, resistor, diode, and inductor. Why?

No. Because diode is a non-linear element.

Question 1 of 2 [8 Marks]

The P vs R graph shows power consumed by a load resistor (R_L).

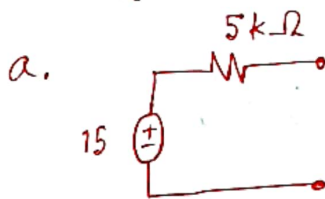
- Draw the Thevenin equivalent circuit that produced the graph.
- Calculate load current I_L for $R_L = R_{Th}$, $R_{Th}/2$, $2R_{Th}$. Which one results in maximum I_L ?

5k ohm, 11.25mW

R_{Th} P_{max}

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

$$\Rightarrow V_{Th} = 15V$$



$$b. I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$R_L = R_{Th} \Rightarrow I_{L1} = \frac{15}{10k} = 1.5mA$$

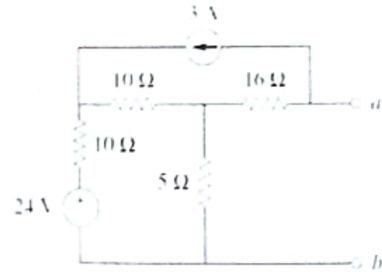
$$R_L = \frac{R_{Th}}{2} \Rightarrow I_{L2} = \frac{15}{7.5} = 2mA$$

$$R_L = 2R_{Th} \Rightarrow I_{L3} = \frac{15}{15} = 1mA$$

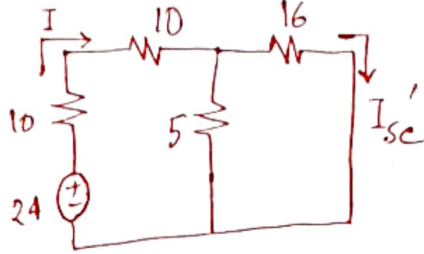
$$\text{max for } R_L = \frac{R_{Th}}{2}$$

Question 2 of 2 [12 Marks]

- a. Use superposition theorem to find the short circuit current between terminal a and b.
b. Draw the Norton equivalent circuit as seen from terminal a and b.



a. 24V active

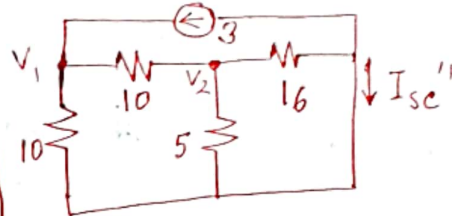


$$I = \frac{24}{20 + (16 \parallel 5)} = 1.008 \text{ A}$$

$$\therefore I_{sc}' = \frac{16 \parallel 5}{16} \times I = 0.24 \text{ A}$$

$$\therefore I_{sc} = (0.24 - 2.7) \text{ A} = \boxed{-2.46 \text{ A}}$$

3A active



$$V_1 \left(\frac{1}{10} + \frac{1}{10} \right) - \frac{V_2}{10} - 3 = 0$$

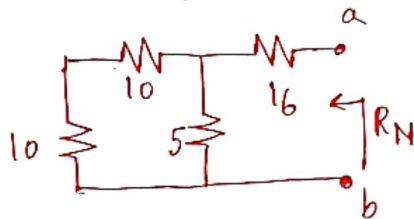
$$V_2 \left(\frac{1}{10} + \frac{1}{5} + \frac{1}{16} \right) - \frac{V_1}{10} = 0$$

$$\Rightarrow V_1 = 17.4 \text{ V}, V_2 = 4.8 \text{ V}$$

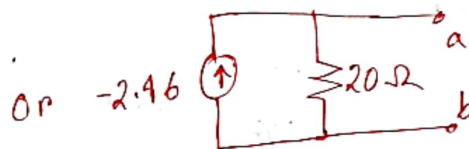
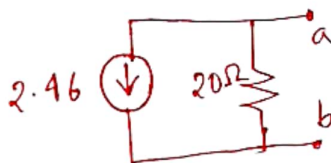
$$3 + I_{sc}'' = \frac{V_2 - 0}{16}$$

$$\Rightarrow I_{sc}'' = -2.7 \text{ A}$$

b. For R_N ,



$$R_N = \{ (10 + 10) \parallel 5 \} + 16 = 20 \Omega$$



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Bonus Question [2 Marks]

Can you apply Thevenin theorem to a circuit that contains the following elements- ac current source, dc current source, resistor, and inductor. Why?

Yes. Because all are linear components.

Question 1 of 2 [8 Marks]

The P vs R graph shows power consumed by a load resistor (R_L).

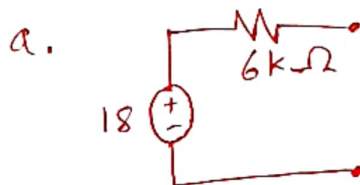
- Draw the Thevenin equivalent circuit that produced the graph.
- Calculate power loss for $R_L = R_{Th}$, $R_{Th}/2$, $2 \cdot R_{Th}$. Which one results in maximum loss?

6k ohm, 13.5mW

R_{Th} P_{max}

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

$$\Rightarrow V_{Th} = 18V$$



b. Power loss = $\cancel{V_{Th}} I^2 R_{Th}$

$$= \frac{V_{Th}^2 R_{Th}}{(R_{Th} + R_L)^2}$$

$$R_{Th} = R_L \Rightarrow P_{loss_1} = 13.5 \text{ mW}$$

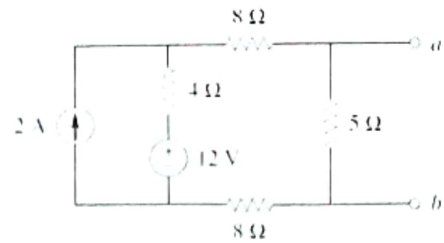
$$\cancel{R_{Th}} R_L = R_{Th}/2 \Rightarrow P_{loss_2} = 2.4 \text{ mW}$$

$$R_L = 2R_{Th} \Rightarrow P_{loss_3} = 6 \text{ mW}$$

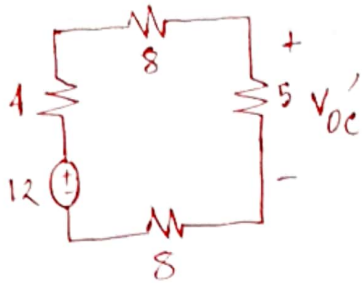
$$\text{max for } R_L = \frac{R_{Th}}{2}$$

Question 2 of 2 [12 Marks]

- a. Use superposition theorem to find the open circuit voltage between terminal a and b.
- b. Draw the Thevenin equivalent circuit as seen from terminal a and b.



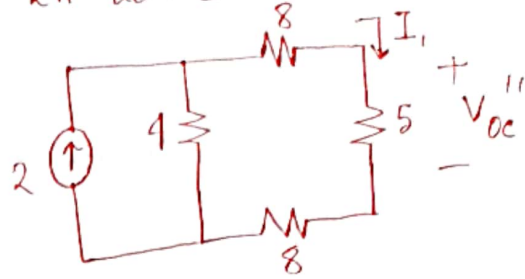
a. 12V active



$$V'_{oc} = \frac{5}{4+8+5+8} \times 12 \text{ V}$$

$$= 2.4 \text{ V}$$

2A active



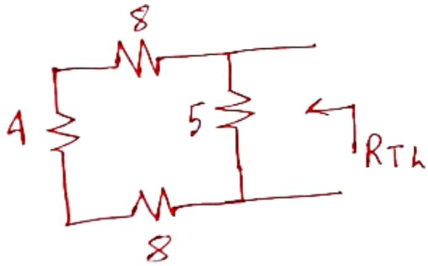
$$I_1 = \frac{2 \parallel 4}{21} \times 2 = 0.32 \text{ A}$$

$$V''_{oc} = I_1 \times 5 = 1.6 \text{ V}$$

$$\therefore V_{oc} = V'_{oc} + V''_{oc}$$

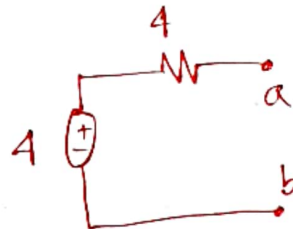
$$= \boxed{4 \text{ V}}$$

b. For R_{Th} ,



$$R_{Th} = (8+4+8) \parallel 5$$

$$= 4 \Omega$$



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Bonus Question [2 Marks]

Can you apply Thevenin theorem to a circuit that contains the following elements- ac voltage source, dc voltage source, resistor, and capacitor. Why?

Yes. Because all are linear components.

Question 1 of 2 [8 Marks]

The P vs R graph shows power consumed by a load resistor (R_L).

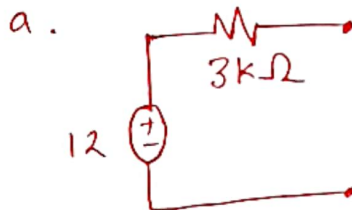
- Draw the Thevenin equivalent circuit that produced the graph.
- Calculate load voltage V_L for $R_L = R_{Th}$, $R_{Th}/2$, $2R_{Th}$. Which one results in maximum V_L ?

3k ohm, 12mW

R_{Th} P_{max}

$$P_{max} = \frac{V_{Th}^2}{4 R_{Th}}$$

$$\Rightarrow V_{Th} = 12V$$



$$b) V_L = \frac{R_L}{R_L + R_{Th}}$$

$$R_L = R_{Th} \Rightarrow V_L = 6V$$

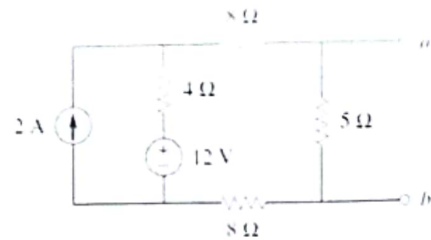
$$R_L = R_{Th}/2 \Rightarrow V_L = 4V$$

$$R_L = 2R_{Th} \Rightarrow V_L = 8V$$

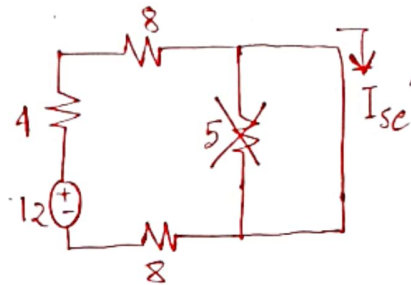
$$\text{max for } R_L = 2R_{Th}$$

Question 2 of 2 [12 Marks]

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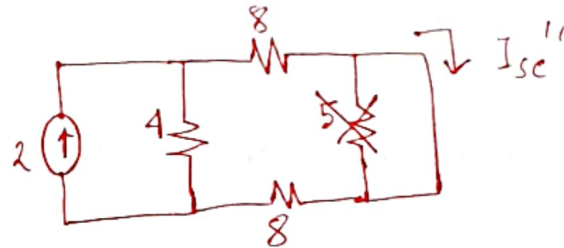


a. 12V active



$$I_{sc}' = \frac{12}{4+8+8} = 0.6A$$

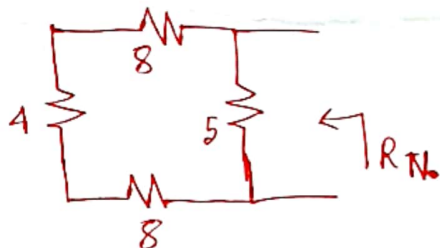
2A active



$$I_{sc}'' = \frac{16 \parallel 4}{16} \times 2 = 0.4A$$

$$\therefore I_{sc} = I_{sc}' + I_{sc}'' = \boxed{1A}$$

b. For R_N ,



$$R_N = (8+4+8) \parallel 5$$

$$= 4\Omega$$

